

SECOND LAW OF THERMODYNAMICS.

(19)

Introduction:

The first law of thermodynamics states that one form of energy can change into another form but the total amount of energy remains the same. Once we specify a particular process or change, this law helps us to balance the internal energy, heat released and work done in the process. But the first law says nothing about whether the process of change we specify can, in fact, occur or to in which direction. This question concerns the second law of thermodynamics. Before the second law, we look at the following definitions of terms.

SPONTANEOUS PROCESSES.

A process which proceeds of its own accord, without any outside assistance, is termed a spontaneous or natural process. The reverse process which does not proceed on its own is referred to as a nonspontaneous or unnatural process. In general, the tendency of a process to occur naturally is called the spontaneity.
Examples: - Rolling ball, Heat flow, Gas flow, Gas mixing.

Criteria of Spontaneity.

Some important criteria of spontaneous physical and chemical change are listed below:-

(1) A spontaneous change is one-way or unidirectional. For reverse change to occur, work has to be done.

- (2) For a spontaneous change to occur, time is not a factor. A spontaneous reaction may take place rapidly or very slowly.
- (3) If a system is not in equilibrium state (unstable), a spontaneous change is inevitable. The change will continue till the system attains the state of equilibrium.
- (4) Once a system is in equilibrium state, it does not undergo any further spontaneous change in state if left undisturbed. To take the system away from equilibrium, some external work must be done on the system.
- (5) A spontaneous change is accompanied by decrease of internal energy or enthalpy (ΔH). It implies that only such reactions will occur which are exothermic. But melting of ice and vaporization of rain water are endothermic processes which proceed spontaneously.
- Clearly, there is some other factor in addition to ΔH which governs spontaneity. It is the second law of thermodynamics which introduces this new factor that is called entropy.

ENTROPY (S)

Entropy is a thermodynamic state quantity that is a measure of the randomness or disorder of the molecules of the system. The symbol of entropy is S , while the change in disorder accompanying a process from start to completion is represented by ΔS . The entropy of a system is a state function and depends only on the initial and final state of the system.

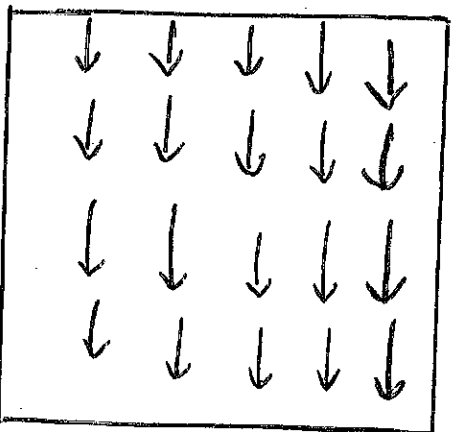
The change in entropy, ΔS , for any process is given by the equation;

$$\Delta S = S_{\text{final}} - S_{\text{initial}}$$

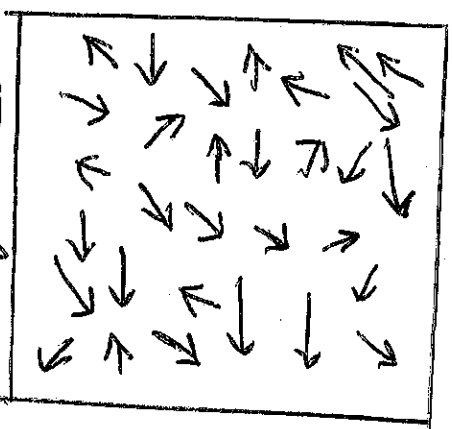
When $S_{\text{final}} > S_{\text{initial}}$, ΔS is +ve.

A process accompanied by an increase in entropy tends to be spontaneous.

Let us consider a molecular system in states A and B. In A, all the molecules are arranged and highly ordered, while in state B, the molecules are present at random and it is highly disordered.



Spontaneous
Change
↓



STATE A
Highly ordered—low
entropy (less probable)

STATE B
Highly disordered—high
entropy (more probable).

(1) By definition, the entropy of A is low and that of B is high. Thus an increase of entropy occurs in the change from A to B.

(2) According to the 2nd law of chance, A is less probable and B is more probable. Therefore the change from A to B is spontaneous.

(3) From (1) and (2), it follows that the change from A to B which is accompanied by increase of entropy will tend to be spontaneous.

Hence, in general, a change in a system which is accompanied by an increase in entropy, tends to be spontaneous.

STATEMENT OF THE SECOND LAW.

The second law of thermodynamics states that: Whenever a spontaneous process takes place, it is accompanied by an increase in ~~entropy~~, the total entropy of the universe.

Universe, means the system and surroundings. Thus,

$$\Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{sur}}$$

The second law stated above tells that when an irreversible spontaneous process occurs, the entropy of the system and the surroundings increases. In other words, $\Delta S_{\text{univ}} > 0$. When a reversible process occurs, the entropy of the system remains constant, $\Delta S_{\text{univ}} = 0$. Since the entire universe is undergoing spontaneous change the second law can be most generally and concisely stated as: The entropy of the system is constantly increasing.

More precisely the term entropy can be defined as: For a reversible change taking place at a fixed temperature (T), the change in entropy ΔS is equal to heat energy absorbed or evolved divided by temperature (T).

(21)

That is,

$$\Delta S = \frac{Q}{T}$$

If heat is absorbed, then ΔS is +ve and there will be increase in entropy. If heat is evolved, ΔS is -ve and there is a decrease in entropy. The unit of entropy is measured in entropy units (eu) which are calories per degree per mole i.e. $\text{cal mol}^{-1} \text{K}^{-1}$ or in SI units ~~are~~ are in Joules per mole per degree i.e. $\text{J mol}^{-1} \text{K}^{-1}$.

Entropy Change for an Ideal gas.

Entropy is a state function and its value depends on two of the three variables T , P and V .

(a) T and V as variables: Let us consider n moles of an ideal gas occupying volume V at pressure P and Temperature T . If the system absorbs dq_{rev} heat reversibly, then increase in entropy is given by:-

$$dS = \frac{dq_{\text{rev}}}{T} \quad \text{--- (1)}$$

but according to the first law of thermodynamics,

$$dq_{\text{rev}} = dE + PdV \quad \text{--- (2)}$$

Putting (2) into (1), we get.

$$dS = \frac{dE + PdV}{T} \quad \text{--- (2)}$$

$$\text{hert } PV = nRT$$

$$\text{or } P = \frac{nRT}{V}$$

$$\text{Also, } dE = n C_v dT \quad \left(C_v = \frac{dE}{dT} \text{ for } n = 1 \text{ mole} \right)$$

where C_v is the molar heat at constant volume.
Substituting (2a) into (2), we have,

$$dS = \frac{n C_v dT + \left(\frac{P}{T} RT \right) dV}{T}$$

$$\text{or } dS = \frac{n C_v dT}{T} + nR \frac{dV}{V} \quad \text{--- (3)}$$

Integrating (3) between limits $S_1, S_2; T_1, T_2; V_1, V_2$,

$$\int_{S_1}^{S_2} dS = \int_{T_1}^{T_2} n C_v \frac{dT}{T} + \int_{V_1}^{V_2} nR \frac{dV}{V}$$

$$\text{or } \Delta S = S_2 - S_1 = n C_v \ln \frac{T_2}{T_1} + nR \ln \frac{V_2}{V_1} \quad \text{--- (4)}$$

$$\text{or } \Delta S = 2.303 n C_v \log \frac{T_2}{T_1} + 2.303 nR \log \frac{V_2}{V_1} \quad \text{--- (5)}$$

for 1 mole of an ideal gas,

$$\Delta S = 2.303 C_v \log \frac{T_2}{T_1} + 2.303 R \log \frac{V_2}{V_1}$$

(b) P and T as variables: let P_1, P_2 be the initial and final pressure states, then,

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

(22)

$$\text{or } \frac{V_2}{V_1} = \frac{P_1 T_2}{P_2 T_1} \quad \text{--- (**)}$$

Substituting (**) into equation (4), we get.

$$\begin{aligned} \Delta S &= n C_v \ln \frac{T_2}{T_1} + n R \ln \frac{P_1 T_2}{P_2 T_1} \\ &= n C_v \ln \frac{T_2}{T_1} + n R \ln \frac{T_2}{T_1} + n R \ln \frac{P_1}{P_2} \\ &= n (C_v + R) \ln \frac{T_2}{T_1} + n R \ln \frac{P_1}{P_2} \\ &= n C_p \ln \frac{T_2}{T_1} + n R \ln \frac{P_1}{P_2} \quad (\text{where } C_p = C_v + R) \\ \therefore \Delta S &= 2.303 \times n \times C_p \log \frac{T_2}{T_1} + 2.303 \times n \times R \log \frac{P_1}{P_2} \quad \text{--- (7)} \end{aligned}$$

for 1 mole of an ideal gas.

$$\Delta S = 2.303 C_p \log \frac{T_2}{T_1} + 2.303 R \log \frac{P_1}{P_2}$$

At constant temperature for an isothermal process

In this case, $T_1 = T_2$, then equation (5) and (6) reduce to:-

$$\Delta S_{\text{rev}} = 2.303 \times n \times R \log \frac{V_2}{V_1} \quad \text{--- (7)}$$

$$\text{and } \Delta S = 2.303 \times n \times R \log \frac{P_1}{P_2} \quad \text{--- (8)}$$

In an isothermal expansion, $V_2 > V_1$ or $P_1 > P_2$, hence ΔS is +ve whereas in isothermal contraction,

$V_2 < V_1$, or $P_1 < P_2$, ΔS_r is -ve.

Example 1: Calculate the entropy change involved in the thermodynamic expansion of 2 moles of a gas from a Vol. of $5L$ to a Vol. of $50L$ at $303K$.

Sol:

Data: $n = 2$; $V_1 = 5L$, $V_2 = 50L$; $R = 8.314J$

Using the relation, $\Delta S_r = 2.303 \times n \times R \log \frac{V_2}{V_1}$

$$\Delta S_r = 2.303 \times 2 \times 8.314 \times \log \frac{50L}{5L} \\ = 38.295 \text{ J K}^{-1}$$

Example 2:

Calculate the entropy change when 2 moles of an ideal gas are allowed to expand isothermally at $293K$ from pressure 10 atm to 2 atm .

Solution:

$$\text{from } \Delta S_r = 2.303 \times n \times R \log \frac{P_1}{P_2}$$

on substitution,

$$\Delta S_r = 2.303 \times 2 \times 8.314 \times \log \frac{10}{2} \\ = 26.77 \text{ J K}^{-1}$$

Entropy Change during Phase Transition

(23)

When there is a change of state from solid to liquid or liquid to vapours or solid to vapours (melting, vaporisation and sublimation respectively), there is a change in entropy. This change may be carried out at constant temperature reversibly or two phases are in equilibrium during the change.

Let us consider the process of melting of 1 mole of the substance being carried out reversibly. It would absorb molar heat of fusion at temperature equal to its melting point. The entropy change is given by:-

$$\Delta S_f = \frac{\Delta H_f}{T_f}$$

where ΔH_f is the molar heat of fusion at its melting point, T_f at constant pressure.

Similarly, when one mole of liquid is boiled reversibly it would absorb molar heat of vaporisation at a temperature equal to its boiling point. In this case, entropy change is given by:-

$$\Delta S_v = \frac{\Delta H_v}{T_b}$$

where ΔH_v is molar heat of vaporisation at its boiling point at constant pressure.

On similar lines, we can calculate the change in entropy when one mole of a solid changes reversibly from one allotropic form to another at its transition temperature. We can write:-

$$\Delta S_f = \frac{\Delta H_f}{T_f}$$

where ΔH_f is the molar heat of transition at its transition temperature T_f .

Example 3: Cal. the entropy change when 1 mole of ethanol is evaporated at 351K = the molar heat of vaporisation of ethanol is 39.84 kJ mol⁻¹.

Solution:

$$\text{from } \Delta S_v = \frac{\Delta H_v}{T_b}, \quad \Delta H_v = 39840 \text{ J mol}^{-1}$$

$$T_b = 351 \text{ K}$$

$$\therefore \Delta S_v = \frac{39840 \text{ J mol}^{-1}}{351 \text{ K}}$$

$$= \underline{\underline{113.53 \text{ J K}^{-1} \text{ mol}^{-1}}}$$

Example 4: 30.4 kJ is required to melt one mole of solid chloride. The entropy change during melting is 28.45 J mol⁻¹ K⁻¹. Cal. the m.p. of solid chloride.

Solution:

$$\Delta S_f = \frac{\Delta H_f}{T_f}$$

$$\Delta S_f = 28.45 \text{ J K}^{-1} \text{ mol}^{-1}; \quad \Delta H_f = 30400 \text{ J}$$

$$\therefore T_f = \frac{\Delta H_f}{\Delta S_f} = \frac{30400 \text{ J}}{28.45 \text{ J K}^{-1} \text{ mol}^{-1}} = \underline{\underline{1070.4 \text{ K}}}$$

24 DETERMINATION OF ENTROPIES OF SUBSTANCES (THE THIRD LAW OF THERMODYNAMICS)

When a system goes from an ordered state to a disordered state, the entropy is increased. The converse is also true. A perfectly ordered crystal at an absolute zero of temperature would have the a zero entropy value since the atoms in a crystal are perfectly ordered and confined to one position only. This gives rise to another empirical law, the third law of thermodynamics, which states:-

"The entropy of a pure perfectly crystalline substance is zero at the absolute zero of temperature."

This statement is applicable only to pure crystalline solid substances because theoretical argument and experimental evidence have shown that the entropy of solutions and supercooled liquids is not zero at OK. For perfectly crystalline substances, the law has been verified repeatedly and at present, there is no doubt about the general validity of the third law.

THE HELMHOLTZ FUNCTION OF FREE ENERGY (A)

This function is defined as $A = E - TS$. ——— ①

To understand the physical significance of A, consider the change in A when the system passes from state 1 to 2.

$$\begin{aligned} \Delta A &= A_2 - A_1 \\ &= (E_2 - T_2 S_2) - (E_1 - T_1 S_1) \end{aligned}$$

$$= E_2 - E_1 - (T_2 S_2 - T_1 S_1)$$

$$\Delta A = \Delta E - (T_2 S_2 - T_1 S_1)$$

Under isothermal condition, $T_2 = T_1 = T$. So that

$$\Delta A = \Delta E - T \Delta S \quad \text{--- (2)}$$

further from entropy, $\Delta S = \frac{q_r}{T}$

or $q_r = T \Delta S$, we have,

$$\Delta A = \Delta E - q_r \quad \text{--- (3)}$$

but since from first law of thermodynamics, for a reversible isothermal process, $\Delta E - q_r = -W_{\max}$, we have,

$$\Delta A = -W_{\max} \quad \text{--- (4)}$$

Hence at constant temperature, the maximum work obtainable from a system is at the expense of a decrease in the Helmholtz free energy of the system. This is why A is sometimes called the "work function" or the maximum content of a system.

Further, if we completely differentiate the equation $A = E - TS$, we get,

$$dA = dE - T dS - S dT \quad \text{--- (5)}$$

But $T dS = q_r$ and $q_r = dE + P dV$, so equation (5) becomes,

$$dA = dE - S dT - P dV \quad \text{--- (6)}$$

At constant volume, $dV = 0$,

$$\left(\frac{\partial A}{\partial T} \right)_V = -S \quad \text{--- (7)}$$

At constant temperature, $dT=0$,

$$\left(\frac{\partial A}{\partial V}\right)_T = -P \quad \text{---} \quad (8)$$

Isobaric change in work function:

for an isothermal change, $dT=0$, and hence equation (6) yields:

$$dA = -P dV$$

for 1 mol of a perfect gas, P may be replaced by $\frac{RT}{V}$, so that

$$dA = -RT \frac{dV}{V} \quad \text{---} \quad (9)$$

for an isobaric process, the increase ΔA in the work function can be obtained by integration of equation (9) b/w limits of the initial and final states.

$$\int_{A_1}^{A_2} dA = - \int_{V_1}^{V_2} RT \frac{dV}{V}$$

$$A_2 - A_1 = -RT \ln \frac{V_2}{V_1}$$

$$\Delta A = -RT \ln \frac{V_2}{V_1} \quad \text{---} \quad (10)$$

THE GIBBS FUNCTION (G)

This can be defined by

$$G = E - TS + PV \quad \text{---} \quad (11)$$

This definition may be written in two alternative but equivalent forms which are frequently employed. First, since $H = E + PV$, then.

$$G = H - TS \quad \text{---} \quad (12)$$

Second, since $A = E - TS$, it follows that,

$$G = A + PV = \quad \quad \quad (13)$$

for a process taking place at constant pressure the change in free energy is given by:

$$\Delta G = \Delta A + P \Delta V = \quad \quad \quad (14)$$

If in addition, the temperature is constant, $\Delta A = -W_{max}$ as seen from equation (4), so that (14) becomes;

$$\Delta G = -(W_{max} - P \Delta V) = \quad \quad \quad (15)$$

The quantity W_{max} represents the maximum work obtainable in the given change and includes all types of work such as electrical or friction work, in addition to work of expansion. The latter is equal to $P \Delta V$ and so $W_{max} - P \Delta V$ represents the reversible work exclusive of expansion, that can be obtained from a given change in state. The quantity $W_{max} - P \Delta V$ is referred to as network

Thus $-\Delta G = \text{network}$.

The decrease in free energy at constant T and P is equal to the maximum network available for the given change in state whilst the process is accompanied.

Again, from the complete differential of equation

$$(12), \text{ we get, } dG = dH - TdS - SdT = \quad \quad \quad (16)$$

(26)

But $H = E + PV$, which on differentiation becomes

$$dH = dE + PdV + VdP, \quad \text{--- (16a)}$$

$$\text{Also, } Tds = dE + PdV \quad \text{--- (16b)}$$

Putting equations (16a and 16b) into (16) we get

$$dG = dE + PdV + VdP - dE - PdV - SdT$$

$$dG = -SdT + VdP \quad \text{--- (17)}$$

At constant pressure, $dP = 0$,

$$\left(\frac{\partial G}{\partial T}\right)_P = -S \quad \text{--- (18)}$$

At constant temperature $dT = 0$,

$$\left(\frac{\partial G}{\partial P}\right)_T = V \quad \text{--- (19)}$$

Isenthal changes in free energy function

At constant temperature, $dT = 0$ and equation (17) becomes,

$$dG = VdP \quad \text{--- (20)}$$

for 1 mole of a perfect gas, $PV = RT$

$$\Rightarrow V = \frac{RT}{P}$$

the equation (20) becomes,

$$dG = \frac{RT}{P} dP \quad \text{--- (21)}$$

Integrating b/w limits, equation (21) becomes

$$\int_{G_1}^{G_2} dG = RT \int_{P_1}^{P_2} \frac{dP}{P} \quad (22)$$

$$G_2 - G_1 = RT \ln \frac{P_2}{P_1} \quad (23)$$

$$\text{or } \Delta G = RT \ln \frac{P_2}{P_1} \quad (24)$$

Generally, the free energy of a gas is related to the standard free energy G° . This is defined as the free energy of one mole of the gas at one atmosphere pressure. Equation (24), then becomes:

$$G - G^\circ = RT \ln \frac{P}{P^\circ} = RT \ln P$$

$$G = G^\circ + RT \ln P \quad (25)$$

THE GIBBS-HELMHOLTZ EQUATION

$$\text{Hence, } G = H - TS$$

Substituting the value of S from equation (26) into equation above, we get

$$G = H - T \left(\frac{\partial G}{\partial T} \right)_P \quad (26)$$

Similarly, substituting the value of S from $\left(\frac{\partial A}{\partial T} \right)_V = -S$ into $A = E - TS$, we get

$$A = E - T \left(\frac{\partial A}{\partial T} \right)_V \quad (27)$$

These two quantities (26 and 27) are usually regarded as different forms of the Gibbs-Helmholtz equation. There are also other forms of the Gibbs-Helmholtz equation which are not widely used, as deduced below:

(27)

For a given reaction of the type,

Reactants $\rightarrow$ Products

$$\Delta G = G_p - G_r$$

where G_p is the sum total of the free energies of all the products and G_r is the sum of the free energies of all the reactants.

From the equation ($G = H - TS$), we have

$G_p = H_p - TS$ and $G_r = H_r - TS$, so that by subtraction

$$\Delta G = (H_p - TS_p) - (H_r - TS_r) \\ = (H_p - H_r) - T(G_p - G_r).$$

$$\text{or } \Delta G = \Delta H - T \Delta S \quad \text{---} \quad (28)$$

where ΔS and ΔH are respectively the entropy and enthalpy change of the given balanced reaction, further, from equation $\left(\frac{\partial G}{\partial T}\right)_p = -S$

$$-\Delta S = -(S_p - S_r) = \left(\frac{\partial G_p}{\partial T}\right)_p - \left(\frac{\partial G_r}{\partial T}\right)_p \\ \text{or } -\Delta S = \left[\frac{\partial (G_p - G_r)}{\partial T} \right]_p = \left[\frac{\partial (\Delta G)}{\partial T} \right]_p \quad \text{---} \quad (29)$$

Substituting this result into equation (28), we have

$$\Delta G = \Delta H + T \left[\frac{\partial (\Delta G)}{\partial T} \right]_p \quad \text{---} \quad (30)$$

Similarly, we can deduce the analogous expression,

$$\Delta A = \Delta E + T \left[\frac{\partial (\Delta A)}{\partial T} \right]_p \quad \text{---} \quad (31)$$

Equations (30) and (31) are the alternate forms of Gibbs-Helmholtz equation.

Example 1: The mole of a perfect gas at 27°C expands isothermally and reversibly from 10 atm to 1 atm against a pressure that is gradually reduced. Cal. q and w . ΔH , ΔA and q ΔS .

Solution:

Since the process is carried out isothermally and reversibly from 10 to 1 atm against a pressure that is reduced.

$$\begin{aligned} W_{\text{max}} &= -RT \ln \frac{V_2}{V_1} = -2.303 RT \log \frac{V_2}{V_1} \\ &= -2.303 RT \log \frac{P_1}{P_2} \\ &= -2.303 \times 8.314 \text{ J mol}^{-1} \text{K}^{-1} \times (300\text{K}) \times \log \frac{10}{1} \\ &= -5744 \text{ J mol}^{-1} \end{aligned}$$

$$\Delta A = W_{\text{max}} = -5744 \text{ J mol}^{-1}$$

Since the internal energy of a perfect gas is not affected by a change in volume, $\Delta E = 0$.

$$\Delta q = \Delta E - W = 0 + 5744 = 5744 \text{ J mol}^{-1}$$

$$\Delta H = \Delta E + \Delta(PV) = 0 + 0 = 0$$

Since PV is constant for a perfect gas at constant temperature,

$$\begin{aligned} \Delta G &= \int_0^1 V dP = RT \ln \frac{10}{1} = 2.303 RT \log \frac{10}{1} \\ &= -2.303 (8.314 \text{ J K}^{-1} \text{ mol}^{-1}) (300\text{K}) \\ &= -5744 \text{ J mol}^{-1} \end{aligned}$$

$$\Delta S = \frac{q_{\text{rev}}}{T} = \frac{5744 \text{ J mol}^{-1}}{300\text{K}} = 19.15 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$\Delta S = \frac{\Delta H - \Delta G}{T} = \frac{0 - (-5744 \text{ J mol}^{-1})}{300\text{K}} = 19.15 \text{ J K}^{-1} \text{ mol}^{-1}$$